

AI-Powered Educational Chatbot

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1 Abstract

This project presents an AI-powered educational chatbot that leverages semantic similarity search to provide accurate answers from a curated knowledge base. Built with Streamlit and Sentence Transformers, the chatbot offers a modern conversational interface with user and bot message bubbles, typing indicators, and chat history support. The system is designed to assist learners by providing quick, reliable, and context-aware responses.

2 Objectives

- Develop a chatbot capable of answering domain-specific questions using a structured dataset.
- Implement semantic search with embeddings and cosine similarity for accurate retrieval.
- Provide a modern, mobile-friendly chat interface with user and bot message bubbles.
- Support fallback responses when information is not available in the dataset.
- Ensure extensibility for future datasets and features.

3 System Architecture

1. **Dataset:** A CSV file containing questions and answers.
2. **Embeddings:** Pre-computed embeddings stored in a CSV file for efficient similarity search.
3. **Model:** SentenceTransformer (`paraphrase-MiniLM-L3-v2`) for generating query embeddings.
4. **Frontend:** Streamlit-based chat interface with sidebar history and suggested questions.
5. **Fallback:** A default response when no relevant answer is found in the dataset.

4 Features

- User and bot message bubbles for a conversational experience.
- Typing indicator with spinner to simulate real-time interaction.
- Sidebar displaying chat history for quick reference.
- Suggested questions for guided exploration.
- Fallback message: “Sorry, I do not have information related to this question.” when no match is found.

5 Dependencies

The project requires the following Python libraries:

- Streamlit
- Pandas
- NumPy
- Sentence Transformers
- Scikit-learn
- Torch
- Transformers
- Tqdm

6 Installation

```
git clone https://github.com/yourusername/educational-chatbot.git
cd educational-chatbot
pip install -r requirements.txt
streamlit run app.py
```

7 Usage

- Launch the Streamlit app.
- Type a question or select from suggested questions.
- View the chatbot’s response from the dataset.
- If no answer is found, a fallback message is displayed.

8 Evaluation

The chatbot was tested with multiple queries from the dataset to ensure:

- Accurate retrieval of answers.
- Proper fallback handling for unknown queries.
- Smooth user experience with chat bubbles and typing indicators.

9 Future Work

- Integration with multiple datasets and knowledge bases.
- Export chat history as text or PDF.
- Enhanced UI with themes and personalization.
- Deployment on cloud platforms (Streamlit Cloud, Hugging Face Spaces).
- Support for multilingual datasets.

10 Conclusion

This project demonstrates the potential of combining semantic search with a modern chat interface to create an effective educational assistant. With further improvements, the chatbot can evolve into a robust learning companion for diverse domains.